

Isaac Mubiru

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PROFESSIONAL SUMMARY

Full-stack engineer with 5+ years shipping end-to-end product features with TypeScript/React front ends and cloud back ends on GCP and AWS. Led the zero-downtime migration of 20+ production applications from Parse/MongoDB to Firebase/GCP, building a real-time data sync pipeline with Node.js and Pub/Sub and Python tooling for large-scale storage migration — cutting infrastructure costs by \$5,000/month. Deep familiarity with APIs, cloud infrastructure, storage solutions, and system design patterns; now applying the same high-ownership, ship-fast approach to generative AI products.

CORE COMPETENCIES

TypeScript / React	Python (tooling & data migration)	Node.js API Design	Cloud Infrastructure (GCP & AWS)
Storage Solutions (Firestore, S3, GCS)	Real-time Data Pipelines (Pub/Sub)	System Design Patterns	OpenAI API Integration

WORK EXPERIENCE

MTailor Sep 2022 – Present

Software Engineer — US Remote

- Led the end-to-end migration of 20+ applications from Parse/MongoDB to Firebase/GCP with zero downtime, owning system design and cloud infrastructure decisions while reporting directly to the CTO
- Built Python tooling to migrate file storage from Amazon S3 to Google Cloud Storage across production storage solutions
- Implemented a real-time two-way sync pipeline between MongoDB and Firestore using Node.js and Google Pub/Sub for message processing
- Built a 3D visualisation feature using video overlay and the ffmpeg library — a proof-of-concept product feature that increased buyer conversion
- Implemented Express Shipping end to end, generating an additional \$40 in revenue per order
- Saved the company \$5,000/month by migrating all services off AWS

MTailor (Contract) Mar 2022 – Jul 2022

Backend Engineer (Contractor) — US Remote

- Built the core migration script reading from MongoDB and writing to Firestore after complex data-processing logic
- Wrote engineering documentation for the new Firebase SDKs in both Node.js and Python; trained the Ops team on dashboards and backup procedures

Dr Wealth Aug 2021 – Jan 2022

Software Engineer — Singapore Remote

- Built batch jobs querying and processing 2+ million Firestore records to keep customer stock prices current from Morningstar APIs — large-scale dataset processing on a live product
- Extended backend APIs with Firebase Cloud Functions and Express.js; shipped responsive React/Tailwind UI pages for the real-time PWA

Mind2matter Jan 2022 – Jul 2022

Software Engineer — US Remote

- Delivered React UIs from Figma designs under tight timelines and built Node.js/Web3 backends for DeFi applications at a US agency with demanding clients

CodeBits Jan 2020 – Jul 2021

Software Engineer (Team Lead) — Uganda Remote

- Architected a microservices backend for FIDA Uganda's case management app using Apache Kafka, Docker, and Kubernetes, leading a team of 4 developers
- Built a USSD service for legal aid providers communicating with a Node.js backend via gRPC; shipped cross-platform React Native apps (iOS & Android)

PROJECTS

Real-time Pub/Sub Sync Pipeline

Live MongoDB ↔ Firestore two-way sync supporting parallel production traffic during a 20-app migration — zero downtime for end users.

Node.js · Google Pub/Sub · Firestore · MongoDB

Healthcare WhatsApp Chatbot

Integrated OpenAI and Google Vision for prescription OCR delivered through the WhatsApp Business API — generative AI applied to a real user workflow.

Node.js · OpenAI API · Google Vision · WhatsApp Business API

Marketplace Recommendation Engine

Collaborative and content-based filtering engine in MongoDB for service-provider matching.

Node.js · MongoDB

EDUCATION

BSc Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

SKILLS

Proficient: Node.js, TypeScript/JavaScript, React, Python, Firebase, GCP, MongoDB, SQL, HTML, CSS, Git, Linux

Intermediate: Go, AWS, Rust, React Native **Languages:** English, Kiswahili